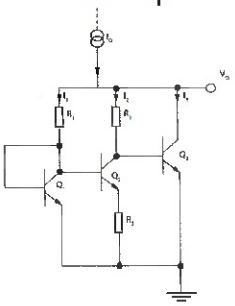
1. (a) Figures 1(a) and 1(b) show two popular biasing schemes typically used in analogue integrated systems. Explain the functionality of each of these circuits, highlighting the differences between the two.

[bookwork]

Band Gap Reference "Diode"



Though the Zener Diode can have a low temperature coefficient, the best "diode" for virtually zero temperature coefficient is the so called **"band-gap diode"**

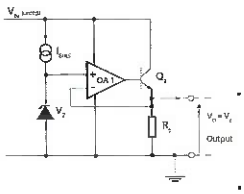
It is a **2 terminal device**, not a single diode at all!

The algebra on the next pages will show that if V_0 is set through the other components to be = **1.283V**, then:

$$\frac{dV_o}{dT} \approx 0 \text{ !!}$$

Such device is very useful in applications where precision is critical such as high resolution ADCs

Practical Circuit of a Zener Reference (3)



- Clearly variations of the input voltage will cause small variations of Zener Voltage, though reduced approximately by the ratio R_2 / R_8
- This is improved by driving the Zener with a perfect DC current source (see next part of this session)
- Also the input impedance and the dependence on the V_{BE} of Q_1 is reduced greatly by using an Op-Amp with negative feedback

[3 points each]

[6]

(b) For the circuit of Figure 1(a), show that $\delta V_0 / \delta T = 0$ (where T is temperature) if $(R_2/R_3)ln[I_1/I_2] = 29$ for $V_0 = 1.283$ V. Assume the temperature coefficient of V_{BE} to be -2.5mV/°C, the collector current of transistor Q_3 is 100μA and the device saturation current is I_S = 1.2 x$10^{-13}$$A$. Boltzmann's constant k = 1.38 x$10^{-23}$$J/K$ and the electron charge is q = 1.6 x$10^{-19}$$C$.

$$-V_{BE1} = V_{BE2} + I_2 R_3 \quad ; \ \beta \rangle\rangle 1$$

$$V_{BE1} - V_{BE2} = V_T \ln\left(I_1/I_2\right)$$

$$V_o = V_{BE3} + \left(R_2/R_3\right) V_T \ln\left(I_1/I_2\right) \quad ; V_{BE3} = V_T \ln\left[I_3/I_s\right] \ assume\ room\ temp$$

$$For \ \partial V_o/\partial T = 0 \Rightarrow \partial V_{BE3}/\partial T = \frac{V_T}{T}\frac{R_2}{R_3}\ln\left[\frac{I_1}{I_2}\right]$$

$$Since \ \partial V_{BE}/\partial T = -2.5\, mV/^{\circ}C \ and \ \frac{V_T}{T} = \frac{1.38\times10^{-23}}{1.6\times10^{-19}}$$

$$\Rightarrow \left(\frac{R_2}{R_3}\right)\ln\left(\frac{I_1}{I_2}\right) = 29 \Rightarrow V_o = 1.283$$

[8]

(c) For the circuit shown in Figure 1(b), show how the op-amp eliminates the voltage error between the Zener voltage and the output load to provide a regulated V_o.

[Derivation from notes]
(Vz-Vout)A=Voa
Voa=Vout+VBE
AVz-AVout=Vout+VBE
Vout(A+1)=AVz+VBE
Assume A>>10^6
Vout=Vz

[3]

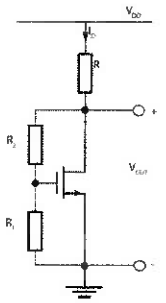
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(d) Sketch the circuit of a VGS multiplier and derive the equation of the output voltage.

V_{GS} Multiplier

Stolen from the V_{BE} multiplier

Circuit Analysis



$$V_{OUT} = V_{GS}\left[1 + \frac{R_2}{R_1}\right]$$

OR

$$V_{OUT} = \left[1 + \frac{R_2}{R_1}\right]\left(V_T + \sqrt{\frac{V_{DD}}{\beta R}}\right)$$

Fig. 16

Note: All R's can be replaced with active resistors

[3]

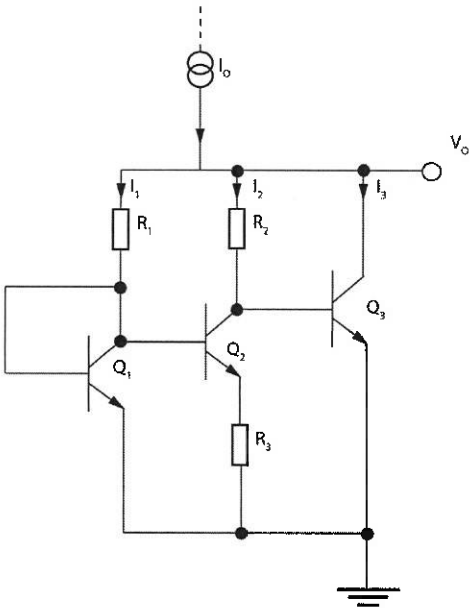


Figure 1(a)

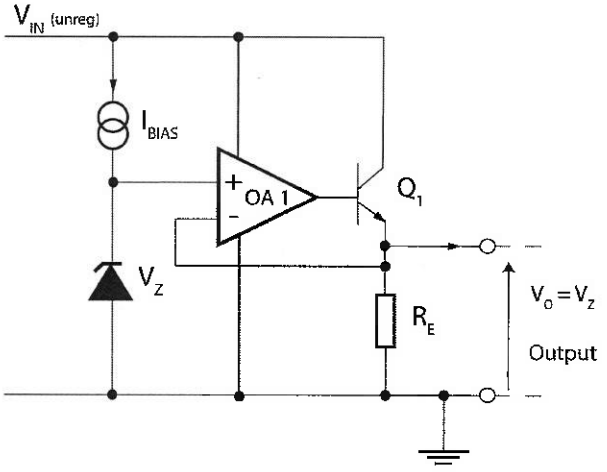
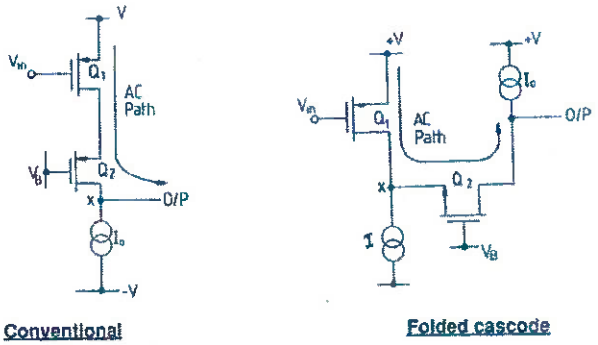


Figure 1(b)

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2. (a) Figure 2(a) shows a folded cascode connection. What is the main advantage of this design over the more classical cascode connection? Show via a brief sketch how the architecture of Figure 2(a) can be used to form the basis of a single stage fully differential folded cascode operational amplifier (op-amp) with common-mode feedback included.

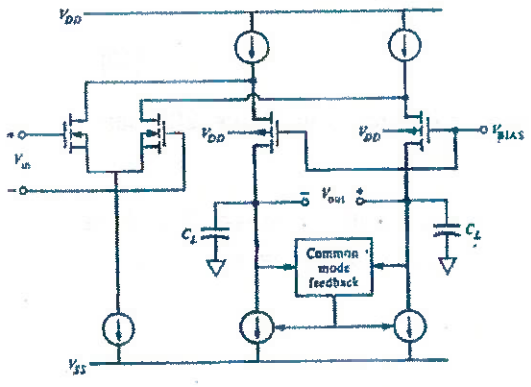
Concept



Conventional 'stacking' cascode

Folded cascode

In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.
Gain of the folded cascode Av=gm1/Go. [3]



[3]

[6]

(b) Figure 2(b) shows a two-stage CMOS op-amp. Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below. What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp?

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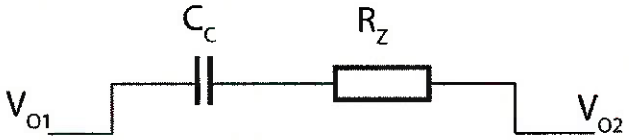
$$Av1 = -gm2/(go2+go4)$$
$$(go2+go4) = I_{D2}(\lambda_N+\lambda_P) = 5\times10^{-6}\times0.05 = 2.5\times10^{-7}\Omega^{-1}$$
$$gm2 = 2\sqrt{\beta_2 I_D}$$
$$\beta_2 = \frac{K_N}{2}\left(\frac{W}{L}\right)_2 = 7.5\times10^{-5}A/V$$
$$gm2 = 3.87\times10^{-5}S$$
$$Av1 = -154.9$$
$$Av2 = -gm6/(go7+go6)$$
$$(go7+go6) = I_{D6}(\lambda_N+\lambda_P) = 20\times10^{-6}\times0.05 = 10\times10^{-7}\Omega^{-1}$$
$$gm6 = 2\sqrt{\beta_6 I_{D6}}$$
$$\beta_6 = \frac{K_N}{2}\left(\frac{W}{L}\right)_6 = 1.6\times10^{-4}A/V$$
$$gm6 = 1.13\times10^{-4}S$$
$$Av2 = -113$$
$$A_{total} = Av1\times Av2 = 17503$$
$$G.Bp = gm2/2\pi C_c = 12.32MHz$$

In a 2-stage opamp the load contributes to the 2nd pole hence reducing load increases stability.
With a single-stage, the load forms the dominant pole hence reducing the load increases bandwidth.

[10]

(c) Explain why a resistor in series with the compensation capacitor *C* in Figure 2(b) can significantly improve the amplifier's phase margin.

Introducing a nulling resistor boot straps the capacitor and significantly increases the phase margin by shifting the second pole. This can also be selected such that the zero introduced cancels out the second pole.



Effectively using a resistor to try and bootstrap C_C, so that at high frequencies it breaks the feed forward path. This is achieved by making
$R_Z = 1/g_{m6}$ so that the voltage across the resistor will cancel the signal voltage on the left hand plate of the capacitor.

The location of the RHP zero however, changes to

$$Z = \frac{1}{C_C\left[\frac{1}{G_{m2}} - R_Z\right]} = \frac{1}{C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

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$$f_Z = \frac{1}{2\pi C_C\left[\frac{1}{g_{m6}} - R_Z\right]}$$

Or

We can cancel Z and push f_Z to infinity by simply making $\boxed{R_Z = \frac{1}{g_{m6}}}$

[4]

CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	*Kp (μA/V²)*	*λ (V⁻¹)*	*V_{T0} (V)*
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

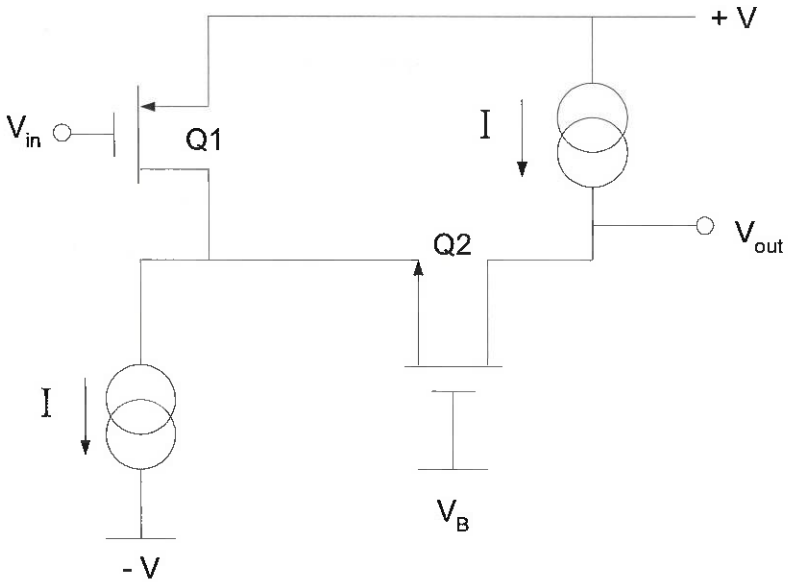


Figure 2(a)

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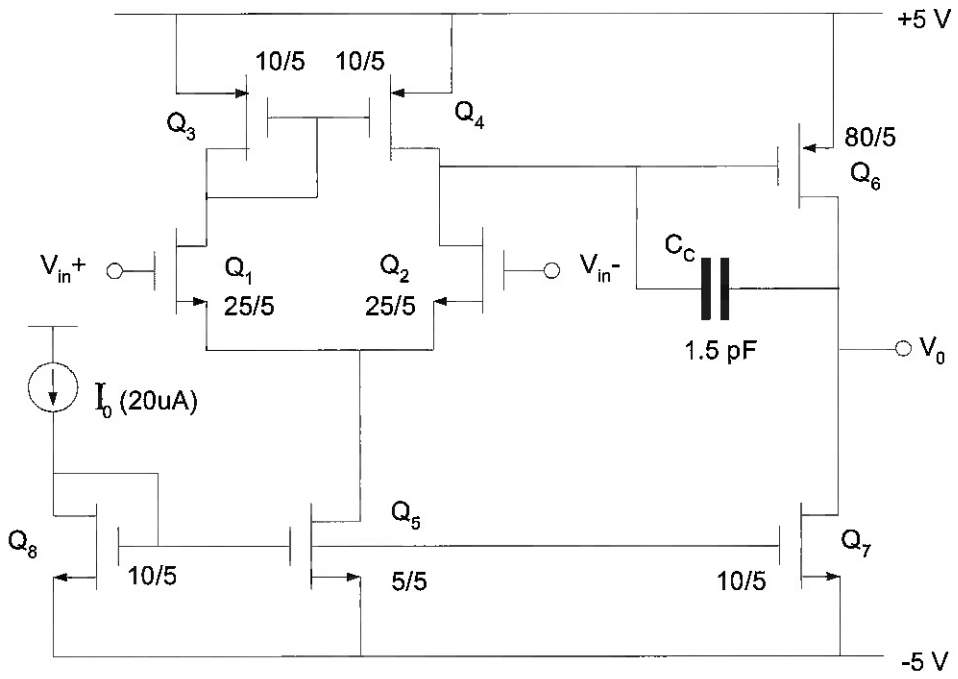


Figure 2(b)

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3. Figures 3(a) and 3(b) show the basic design of two integrated circuit precision integrators.

(a) Derive an expression for the time constant of each integrator. Assume that the sampled-data integrator of Figure 3(b) is driven by non-overlapping clocks and that the switches are ideal.

Floating resistor RC integrator, double MOS differential eliminating Vds and Vt,

$$I_{D1} = 2\beta\left[(V_{C1} - V - V_T)(V_{In} - V) - 1/2(V_{In} - V)^2\right]$$
$$I_{D2} = 2\beta\left[(V_{C2} - V - V_T)(V_{In} - V) - 1/2(V_{In} - V)^2\right]$$
$$I_{D3} = 2\beta\left[(V_{C2} - V - V_T)(-V_{In} - V) - 1/2(-V_{In} - V)^2\right]$$
$$I_{D4} = 2\beta\left[(V_{C2} - V - V_T)(-V_{In} - V) - 1/2(-V_{In} - V)^2\right]$$
$$I_1 = I_{D1} + I_{D3}\ ;\ I_2 = I_{D2} + I_{D4}$$
$$\Rightarrow (V_{In} - (-V_{In}))/(I_1 - I_2) = 1/2\beta(V_{C1} - V_{C2}) = R$$
$$\tau = RC = \frac{C}{2\beta(V_{C1} - V_{C2})}$$

Switched capacitor lossy integrator

Differential Integrator (3)

$$V_O = \frac{1}{jf/f_O}(V_2 - V_1) \qquad f_O = \frac{1}{2\pi}\frac{C_1}{C_2} f_c$$

OPERATION

Φ_1 $\quad Q = C_1(V_1 - V_2)$

$$I_{av} = f_C C_1(V_1 - V_2)$$

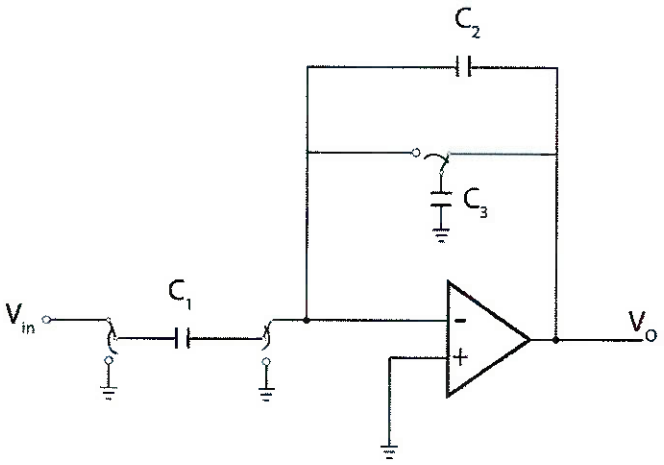
Φ_2 $\quad I_{av[Op-amp]} = -f_C C_1(V_1 - V_2)$

$$V_O = \frac{1}{j\omega C_2} f_C C_1[V_1 - V_2]$$

[5]

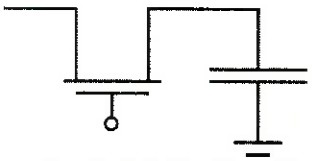
(b) Show how the integrator in Figure 3(b) can be made lossy and sketch the new schematic.

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[4]

(c) Explain what noise source limits the dynamic range of sample data systems.



Sampling Switch Configuration.

$DR \therefore V_{\mathrm{Re}f} / Noise = 2^N$

RMS noise of switch capacitance $\sqrt{\frac{kT}{C}}$

Assume

$f_C = \frac{1}{10.R.C}$ Then solving for C gives

$DR = 2^N = V_{\mathrm{Re}f} / \sqrt{kT\,10Rf_C}$

$R_{On} = \frac{1}{2\beta\left(V_{gs} - V_T\right)} \approx \frac{1}{\left(2\beta \times 4\right)} = 100k\Omega \; ; \beta = \left(\frac{kW}{2L}\right)$

Can now find DR 40kHz,

$DR = 3.88 \times 10^5 \Rightarrow N = 18.5 -> 18$

[3]

(d) The integrator in Figure 3(b) also includes an ADC. Describe the operation and sketch the architecture of a suitable ADC architecture which overcomes this limitation in part 3.c with oversampling.

In the sigma-delta modulator, coarse quantization at high sampling rate is combined with negative feedback and digital filtering to achieve increased resolution at lower sampling rates. The architecture comprises a negative feedback loop producing a coarse estimate that oscillates about the true value of input. The feedback A/D and integrator (SC) force the quantizaion error (noise) to have a high frequency spectrum. The output of the

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modulator is down sampled and gives a multibit digital representation oat lower sampling rate, it averages the coarse quantization leading to a much finer quantization. Since quantization noise is reduced this leads to higher resolution and S/N ratio. When N is 1 the coarse estimate and A/D converter have a 1 bit resolution and this part of the circuit is referred to the sigma-delta modulator. Basically quantisation noise is reduced because the digital filter (decimator) rejects quantisation noise above the signal. Over sampling stretches noise but keeps the signal the same. Generally quantisation noise is flat over the oversampled frequency range. Higher order sigma-delta modulators shape the quantisation noise (eg low at low frequency and high at high frequency) hence filtering will reject a higher proportion.

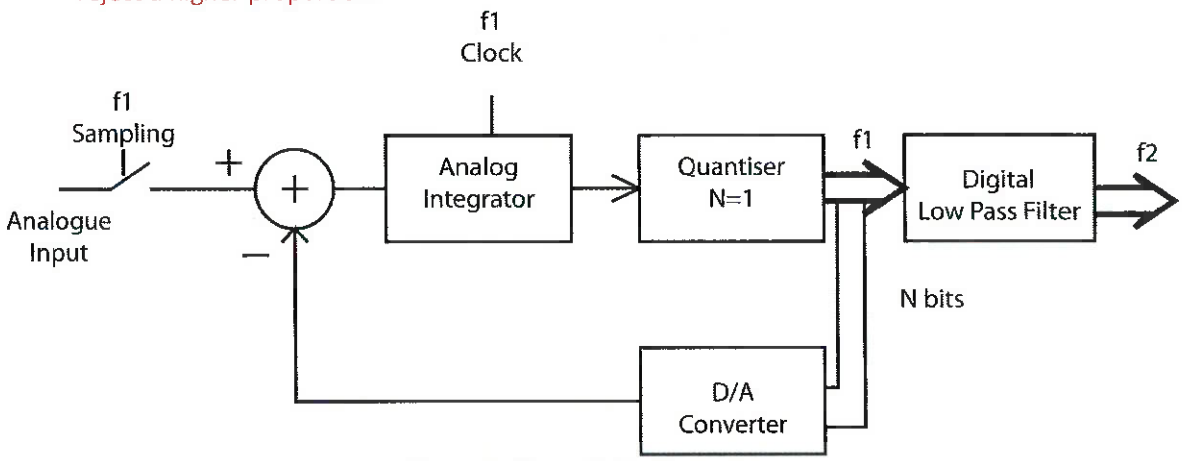


Figure 1. Sigma Delta Converter.

[8]

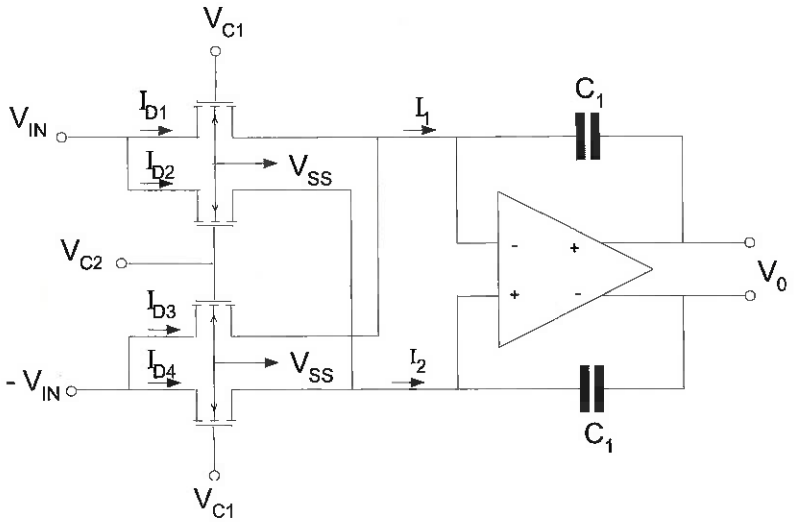


Figure 3(a)

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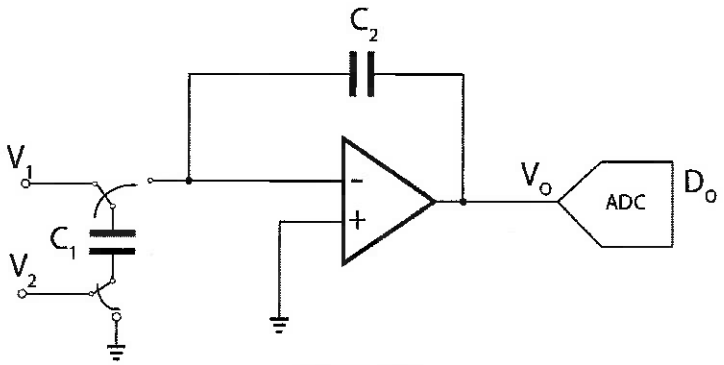
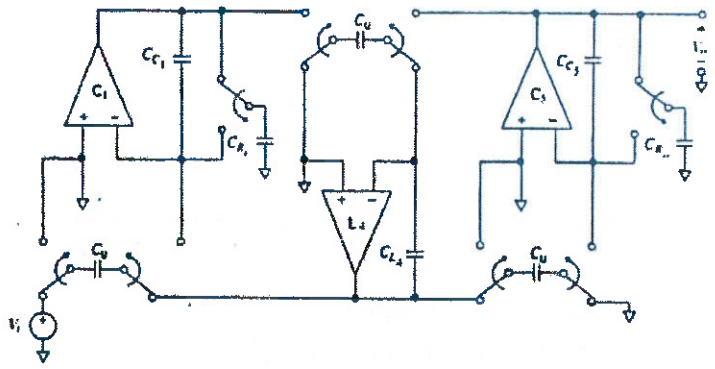


Figure 3(b)

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4.

(a) Sketch the basic design of a 3rd-order Chebyshev low pass switched-capacitor ladder filter.



[5]

The filter is to have a cut-off frequency of 10 kHz and assume a clocking frequency of 200 kHz. The values of the integration capacitor for the capacitor-based sections are 10.12 pF, and for the inductive based sections are 6.98 pF. All other switched capacitors are 1pF.

(b) From the circuit, estimate normalised passive component values for the original double-terminated LC prototype of the filter. All values should be normalised to 1 rad/s. You may assume that the clocking frequency is so high that the integrators can be assumed lossless.

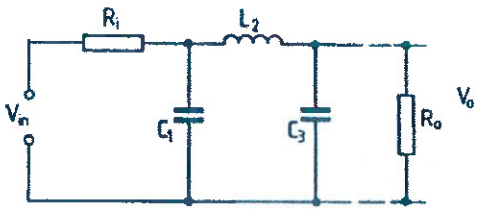
a. General transformation rules for ladder prototypes:

Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype



For switched capacitor equivalent:
C_{c1}=Cc3=10.12 pF
C_{L2}=6.98 pF.
C_u=1pF
Assume scaling Rs=Ri=Ro=1Ω

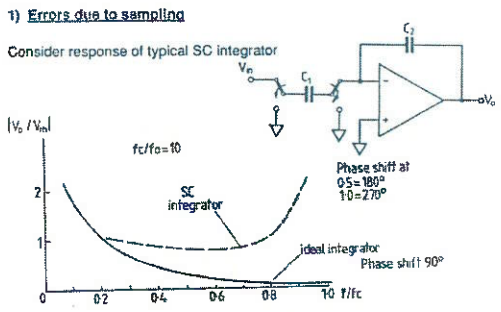
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Therefore
$L2 = CL2/fc = 6.98/200\times10^{3} = 3.49\times10^{-5}$ H

Normalised 1rad/sec we multiply by $2\pi f_0$
$L2 = 3.49\times10^{-5} \times 2\pi \times 10$ KHz $= 2.192$ H
$C1 = C3 = Cc3/fc = 10.12/200\times10^{3} = 5.08\times10^{-5}$ f
Normalised value C1=C3=3.192 f

[8]

(c) Explain the limitations of switch capacitor filters, showing a graph and a transfer function of how it's response deviates from an ideal integrator. Explain how this imposes a limitation on the clock frequency.



Exact expression for SC integrator

$$V_o/V_{in} = \frac{1}{jf/fo} \cdot \left[\frac{\omega/fc}{2\sin\omega/2fc}\right] \exp(-j\omega/2fc)$$

IDEAL INTEGRATOR — AMPLITUDE ERROR — PHASE ERROR

For f/fc → 0 where $\omega = 2\Pi f$

$V_o/V_{in} = 1/jf/fo$ response ideal and continuous.

For fc > f descrete time - non-ideal.

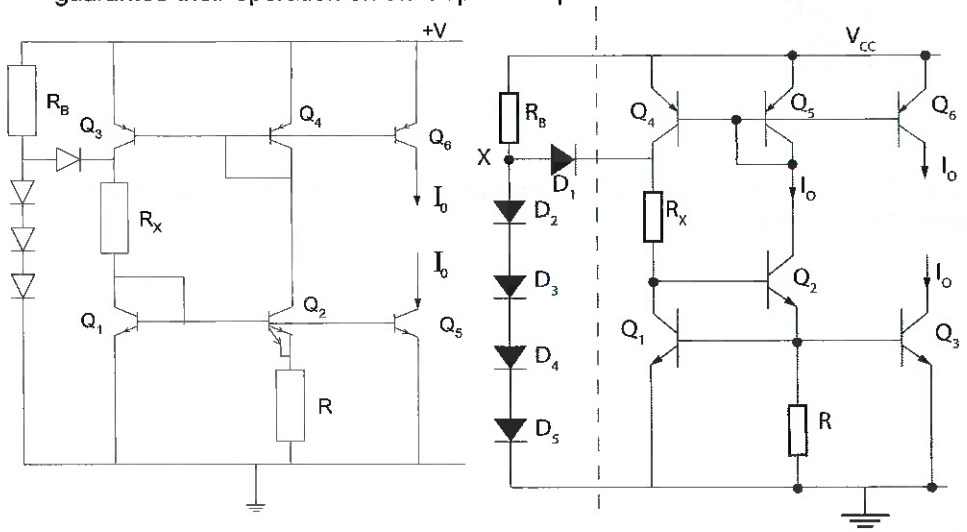
[5]

(d) Explain how the errors due to the limitations explained in part (c) be reduced.
Antialiasing of smoothing filters at the output may be used to filter out the sampling components.

[2]

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5. (a) Sketch the circuits of two supply independent current sources, including circuitry to guarantee their operation on start-up and explain the main differences.



[6]

(b) Explain the necessity of the automatic start up circuit for these topologies deriving the possible biasing points which exist.

Student must explain they are self-biased current references and explain the need and role of the startup circuit based on equations. [4 points]

Equation for output current:

$$I_O = \frac{(V_{be1} - V_{be2})}{R} = \frac{V_T \ln\left[\frac{I_{IN}}{I_O}\right]}{R} = \frac{V_T}{R} \ln\left[\frac{I_{IN}}{I_{S1}} \times \frac{I_{S2}}{I_O}\right]$$

If we self bias and force $I_{IN}/I_O = 1$ with top current mirror, then ln(1) = 0
Require $I_{IN}/I_O > 1$
In Figure 1.1, $I_{IN}/I_O = n = 2$
So, assuming matched devices

$$I_O = \frac{V_T}{R} \ln[2]$$

[5]

(c) Using the equation for fractional temperature coefficient, explain which of the two circuits in 5.a is more insensitive to temperature.

Must elaborate that one circuit has a better temperature coefficient that that the other and show equations for Temperature coefficient.
Elimination of Vbe produces an Improved TCF using VT Referenced Current Source/Sink

$$TC_F = \left(\frac{1}{V_T}\right)\frac{\delta V_T}{\delta T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$

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A V_{be} Referenced Sink/Source with a temperature coefficient has worse performance:

$$TC_F = \left(\frac{1}{V_{be1}}\right)\frac{\delta V_{be1}}{\delta T} - \left(\frac{1}{R}\right)\frac{\delta R}{\delta T}$$

[4]

(d) Describe how a PTAT could be used to bias long-tail pair such that it's transconductance becomes independent of temperature.

$$GBW = \frac{g_{m2}}{2\pi C_c}$$

$$g_m = \frac{I_0}{V_t}$$

$$I_0 = (V_{be1} - V_{be2})/R = V_T \ln[I_{in}/I_0]/R$$

$$I_{in}/I_0 = 2$$

$$g_m = \frac{V_T \ln[2]/R}{2\pi V_t} = \frac{\ln[2]}{2\pi R}$$

$$GBW = \frac{\ln[2]}{2\pi R C_C}$$

[5]

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6. (a) Give one advantage and one disadvantage for each of the two CMOS current mirror circuits shown in Figures 6.1 and 6.2.

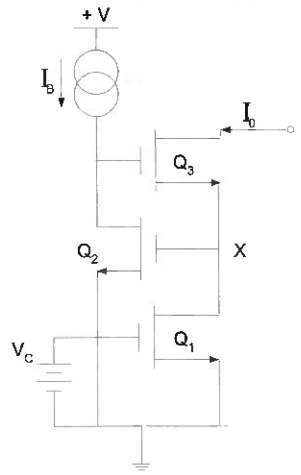
Figure 6.1
Advantage: Large voltage headroom/good for low voltage operation as has needs only one ΔV on output to be in saturation.
Disadvantage: Errors on output current due to ΔVDS will be significant.

Figure 6.2
Advantage: Improved output impedance due to cascoding and therefore less error due to early effect, changes in ΔVDS
Disadvantage: Two output transistors reduce output voltage headroom because circuit needs two ΔV to keep output transistors in saturation.

[4]

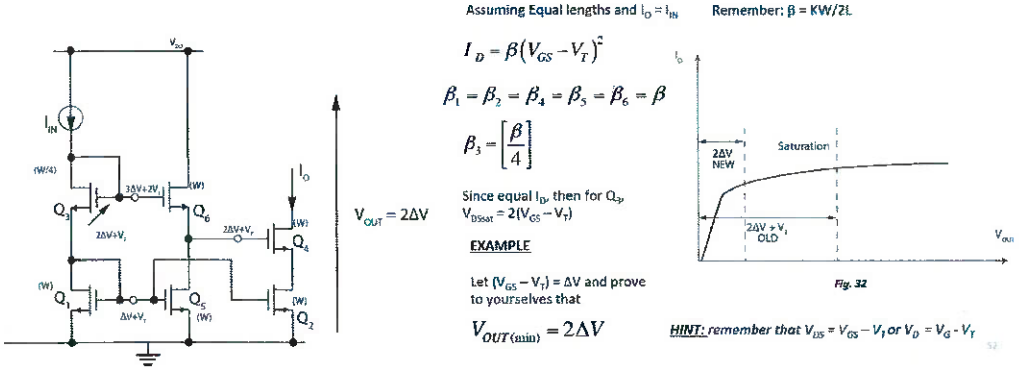
(b) For the current mirror of Figure 6.1, show how using a regulated cascode you can improve the output impedance stating equations for the output impedance before and after the improvement.

[bookwork and derivation from notes of regulated cascode]



[6]

(c) For the current mirror of Figure 6.2 derive this voltage swing in terms of device threshold voltage V_T, clearly stating any assumptions you make.

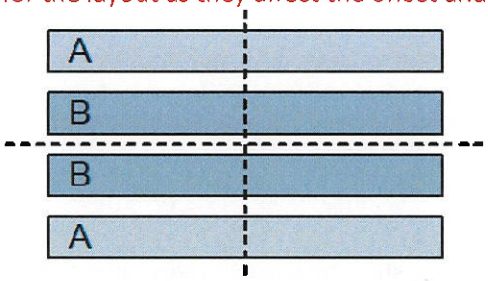


[5]

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(d) Discuss a suitable layout technique which would ensure the transistors in your current mirror are matched.

Mismatches like λ (the Vds difference), threshold voltage VT and W/L may cause errors in offset and gain. A proper layout may help reduce the effect of such mismatches as well as gradients. Common centroid design splits the transistors over a common centroid or axis of symmetry. For example, the elements A and B in an array may be inter-digitated as in figure below. The input differential pair transistors in the Op-amp are the critical ones for the layout as they affect the offset and gain.



interdigitation over a common centroid.

Description of common centroid concept and telling the crcitical transistors of the Op-Amp for the layout

[5]

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